

Chapter 6: Sampling Distributions

Probability and Statistics for Science and Engineering
with Examples in R

Hongshik Ahn

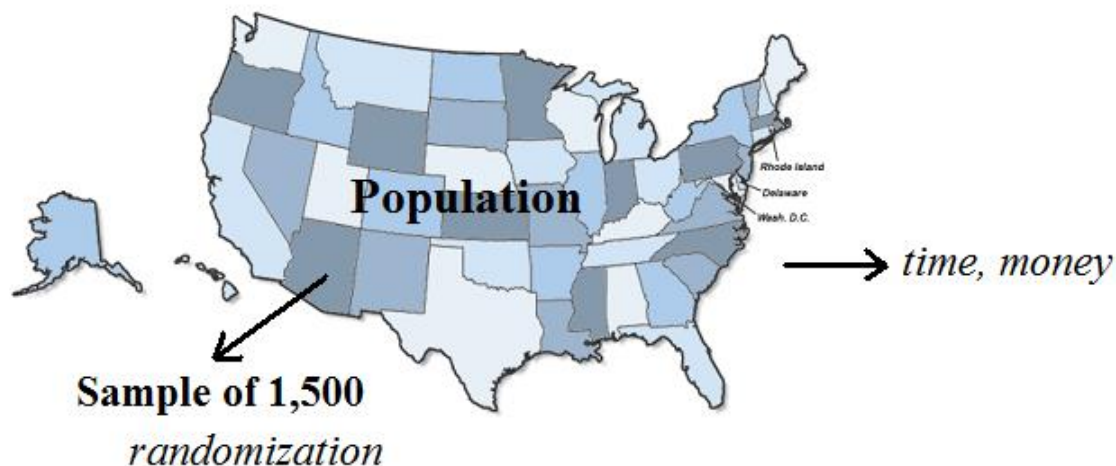
(1) Populations & Samples

- Population: The vast collection of all potential observations of a particular type. eg) enrollment figure, height, ...
- Sample: The data set that is actually acquired through the process of observation

Example 6.1:

(a) Election poll

600 of them
support candidate A.



Support rate of candidate A:

$$p = 0.4 \Rightarrow \text{sd} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.4)(0.6)}{1500}} \approx .0126$$
$$95\% \text{ CI: } p \pm 1.96\text{sd} = (0.375, 0.425)$$

(b) Collision test of cars (exhaustive set of data is not available)

Major Objectives of Statistics

1. Make inferences about a population from an analysis of information contained in sample data
2. Design the process & the extent of sampling so that the observations form a basis for drawing valid inferences

- The r.v.s $X_1, \dots, X_n$ are a random sample of size n (iid) if
 - 1) $X_1, \dots, X_n$ are independent
 - 2) Every X_i has the same distribution(iid: independently and identically distributed)

- Parameter: population quantity ($\mu, \sigma^2, \dots$)
- Statistic: A function of one or more r.v.s that does not depend on any unknown parameter (eg. $\bar{X}, \sum X_i, S^2, \dots$)

(2) Sampling Distribution of the Mean

Let $X_1, \dots, X_n$ be a random sample from a distribution with mean μ & standard deviation σ . Then

$$1. E(\bar{X}) = \mu$$

$$2. \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

$$3. E\left(\sum_{i=1}^n X_i\right) = n\mu,$$

$$4. \text{Var}\left(\sum_{i=1}^n X_i\right) = n\sigma^2, \quad \sigma_{\sum X_i} = \sqrt{n}\sigma$$

Example 6.2:

Grades of a nation-wide exam: $\mu = 70$ & $\sigma = 14$

$X_1, X_2, \dots, X_{36}$: random sample

$$E(\bar{X}) = \mu = 70$$

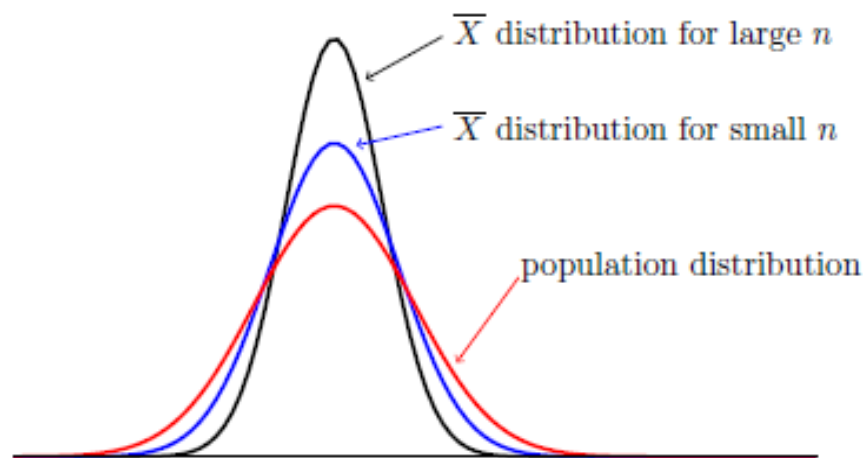
$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{14}{6} = 2.33$$

$$E\left(\sum_{i=1}^{36} X_i\right) = n\mu = 36(70) = 2,520$$

$$\sigma_{\sum X_i} = \sqrt{n}\sigma = 6 \cdot 14 = 84$$

- $\sigma_{\bar{X}}$ decreases as n grows even though $E(\bar{X})$ stays the same. The distribution of $\bar{X}$ tends to become more concentrated around $\bar{X}$. More information (more data) $\Rightarrow$ more certain about the estimation of μ .

$\bar{X}$ is normal when sampling from a normal population.
Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Then $\bar{X} \sim N(\mu, \sigma^2/n)$.



Example 6.3 :

The time that it takes a randomly selected sports car from the same brand to reach 60 miles per hour (mph) from start: $N(6.2, (0.15)^2)$ (in seconds). $X_1, \dots, X_6$: random sample

(a) Probability that the total time $T = \sum_{i=1}^6 X_i$ is between 36.5 and 37.5 seconds?

$$E(T) = E\left(\sum_{i=1}^n X_i\right) = n\mu = 6(6.2) = 37.2$$

$$\sigma_T = \sigma_{\sum_{i=1}^n X_i} = \sqrt{n}\sigma = 0.15\sqrt{6} = 0.367$$

$$T \sim N(37.2, (0.367)^2)$$

$$P(36.5 < T < 37.5) = P\left(\frac{36.5 - 37.2}{0.367} < Z < \frac{37.5 - 37.2}{0.367}\right)$$

$$= P(-1.91 < Z < 0.82)$$

$$= \Phi(0.82) - \Phi(-1.91) = 0.7939 - 0.0281 = 0.7658$$

Example 6.3 (continued) :

$N(6.2, (0.15)^2)$ (in seconds). $X_1, \dots, X_6$: random sample

(b) Probability that the sample average time $\bar{X}$ is at most 6.0 seconds?

$$E(\bar{X}) = \mu = 6.2 \text{ and } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.15}{\sqrt{6}} = 0.0612$$

$$\bar{X} \sim N(6.2, (0.0612)^2)$$

$$P(\bar{X} < 6.0) = P\left(Z < \frac{6.0 - 6.2}{0.0612}\right) = \Phi(-3.27) = 0.0005$$

(3) Central Limit Theorem (CLT)

Let $X_1, \dots, X_n$ be a random sample from any distribution with mean μ & variance σ^2 . Then if n is large, $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is approx. $N(0,1)$.

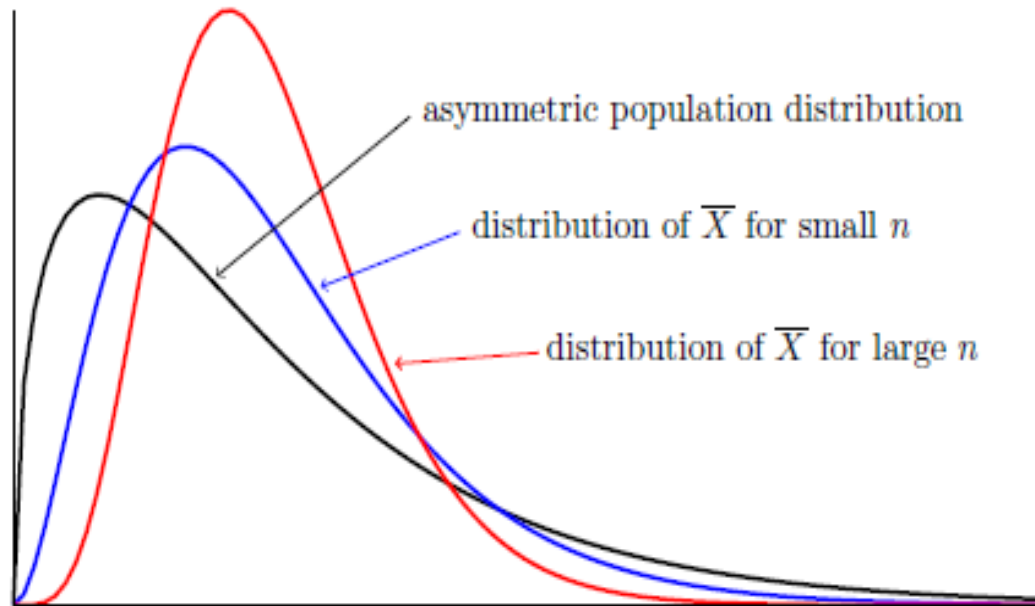
($n \geq 30$: enough)

Example 6.4: Normal approximation to the binomial

$$\begin{aligned} Z &= \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - p}{\sqrt{p(1-p)}/\sqrt{n}} = \frac{n(\bar{X} - p)}{\sqrt{np(1-p)}} = \frac{n\bar{X} - np}{\sqrt{np(1-p)}} \\ &= \frac{Y - np}{\sqrt{np(1-p)}} \rightarrow N(0,1) \\ &\text{as } n \rightarrow \infty \end{aligned}$$

Example 6.5:

Let $X_1, \dots, X_n$ be a random sample. Let $\mu = 2, \sigma = 1.41$, $n = 3$ or $n = 10$.



Example 6.6: $X_1, X_2, \dots, X_n$: random sample, $\mu = 84, \sigma = 14$

(a) When $n = 49$, find $P(82.6 < \bar{X} < 86.8)$.

$$E(\bar{X}) = \mu = 84, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{14}{\sqrt{49}} = 2$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 84}{2} \rightarrow N(0,1)$$

$$\begin{aligned} P(82.6 < \bar{X} < 86.8) &= P\left(\frac{82.6 - 84}{2} < Z < \frac{86.8 - 84}{2}\right) \\ &= P(-.7 < Z < 1.4) \approx \Phi(1.4) - \Phi(-0.7) = 0.9192 - 0.2420 = 0.6772 \end{aligned}$$

(b) When $n = 100$, find $P(82.6 < \bar{X} < 86.8)$.

$$E(\bar{X}) = \mu = 84, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{14}{\sqrt{100}} = 1.4$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 84}{1.4} \rightarrow N(0,1)$$

$$\begin{aligned} P(82.6 < \bar{X} < 86.8) &= P\left(\frac{82.6 - 84}{1.4} < Z < \frac{86.8 - 84}{1.4}\right) = P(-1 < Z < 2) \\ &\approx \Phi(2) - \Phi(-1) = 0.9772 - 0.1587 = 0.8185 \end{aligned}$$

Example 6.7: The weights of 7th grade male students of a certain school have mean 109 pounds and standard deviation 12. Let $X_1, X_2, \dots, X_{64}$ be a random sample from this population. Find an approximate probability that the sample mean exceeds 112 pounds.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - 109}{12/\sqrt{64}} = \frac{\bar{X} - 109}{1.5} \text{ is approximately } N(0,1).$$

$$\begin{aligned} P(\bar{X} > 112) &= P\left(Z > \frac{112 - 109}{1.5}\right) = P(Z > 2) \approx \Phi(-2) \\ &= 0.0228 \end{aligned}$$

(4) Distribution of Sample mean, Unknown σ

If $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$ and $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$, then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

T distribution with df (degrees of freedom) ν (t_ν):

1. Bell-shaped, symmetric about 0
2. More spread out than $N(0,1)$ curve
3. As ν increases, tails become thinner
4. As $\nu \rightarrow \infty$, $t_\nu \rightarrow N(0,1)$ (Approximation is good when $\nu \geq 30$)

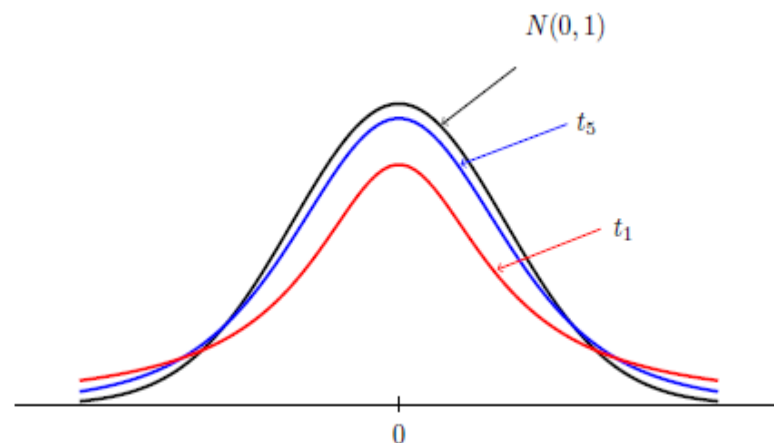
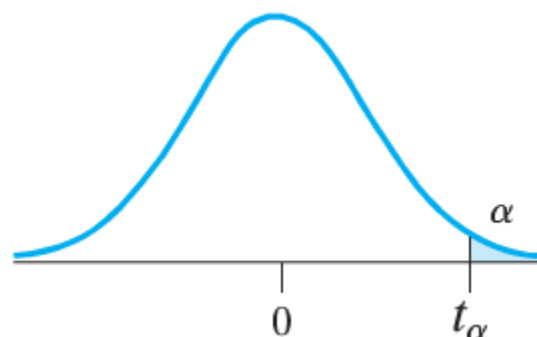


Table A.4

t_α : t critical value

$$p(t_v > t_{\alpha,v}) = \alpha$$



ν	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$	$\alpha = 0.00833$	$\alpha = 0.00625$	$\alpha = 0.005$	ν
1	3.078	6.314	12.706	31.821	38.204	50.923	63.657	1
2	1.886	2.920	4.303	6.965	7.650	8.860	9.925	2
3	1.638	2.353	3.182	4.541	4.857	5.392	5.841	3
4	1.533	2.132	2.776	3.747	3.961	4.315	4.604	4
5	1.476	2.015	2.571	3.365	3.534	3.810	4.032	5
6	1.440	1.943	2.447	3.143	3.288	3.521	3.707	6
7	1.415	1.895	2.365	2.998	3.128	3.335	3.499	7
8	1.397	1.860	2.306	2.896	3.016	3.206	3.355	8
9	1.383	1.833	2.262	2.821	2.934	3.111	3.250	9
10	1.372	1.812	2.228	2.764	2.870	3.038	3.169	10
11	1.363	1.796	2.201	2.718	2.820	2.891	3.106	11
12	1.356	1.782	2.179	2.681	2.780	2.934	3.055	12
13	1.350	1.771	2.160	2.650	2.746	2.896	3.012	13
14	1.345	1.761	2.145	2.624	2.718	2.864	2.977	14
15	1.341	1.753	2.131	2.602	2.694	2.837	2.947	15
16	1.337	1.746	2.120	2.583	2.673	2.813	2.921	16
17	1.333	1.740	2.110	2.567	2.655	2.793	2.898	17
18	1.330	1.734	2.101	2.552	2.639	2.775	2.878	18
19	1.328	1.729	2.093	2.539	2.625	2.759	2.861	19
20	1.325	1.725	2.086	2.528	2.613	2.744	2.845	20
21	1.323	1.721	2.080	2.518	2.602	2.732	2.831	21
22	1.321	1.717	2.074	2.508	2.591	2.720	2.819	22
23	1.319	1.714	2.069	2.500	2.582	2.710	2.807	23
24	1.318	1.711	2.064	2.492	2.574	2.700	2.797	24
25	1.316	1.708	2.060	2.485	2.566	2.692	2.787	25
26	1.315	1.706	2.056	2.479	2.559	2.684	2.779	26
27	1.314	1.703	2.052	2.473	2.553	2.676	2.771	27
28	1.313	1.701	2.048	2.467	2.547	2.669	2.763	28
29	1.311	1.699	2.045	2.462	2.541	2.663	2.756	29
inf.	1.282	1.645	1.960	2.326	2.394	2.498	2.576	inf.

Example 6.8:

The average gasoline consumption of an engine for half-an-hour run is 2.4 gallons. Let $X_1, \dots, X_9$ be gallons used from 9 half-an-hour random test runs. Then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\bar{X} - 2.4}{S/\sqrt{9}} \sim t_{n-1} = t_8$$

Using R,

$P(T < c)$, where $T \sim t_v$:

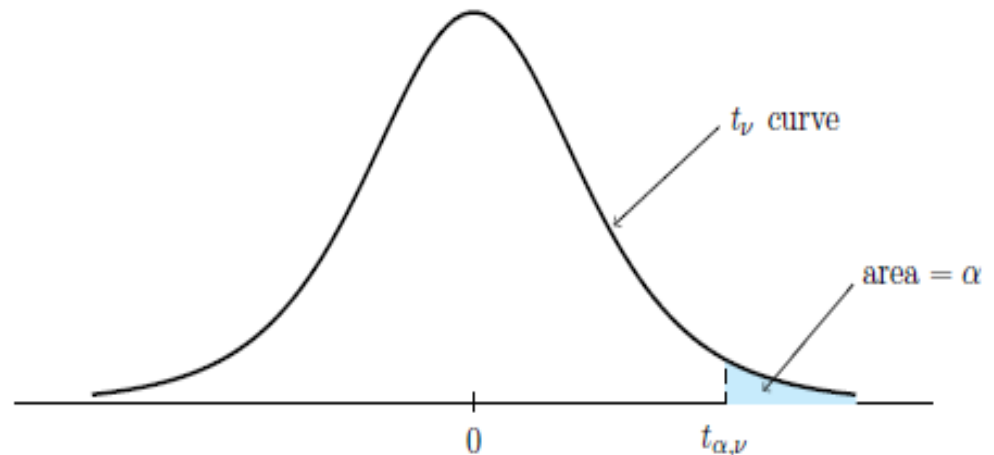
`>pt(c,v)`

100p-th percentile:

`>qt(p,v)`

$t_{\alpha,v}$:

`>qt(1-  $\alpha$ , v)`



Example 6.9:

The times between the call for an ambulance and the patient's arrival at the hospital are 27, 15, 20, 32, 18 and 26 minutes. The ambulance service claims that their average is 20 minutes. In view of the data, is their claim reasonable?

$n = 6$, $\bar{x} = 23$ minutes and $s = 6.39$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{23 - 20}{6.39/\sqrt{6}} = 1.15$$

$P(\bar{X} > 23) = P(T > 1.15) > 0.1$ because $t_{0.1,5} = 1.476$.

Using R,

```
>pt(1.476, 5)  
0.9000149
```

```
>1-pt(1.476,5)  
0.09998513
```

```
P(T > 1.15):  
>1-pt(1.15, 5)  
0.1510794
```

```
>qt(0.9, 5)  
1.475884
```


(5) Sampling Distribution of the Variance

If $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$$

χ_v^2 density functions

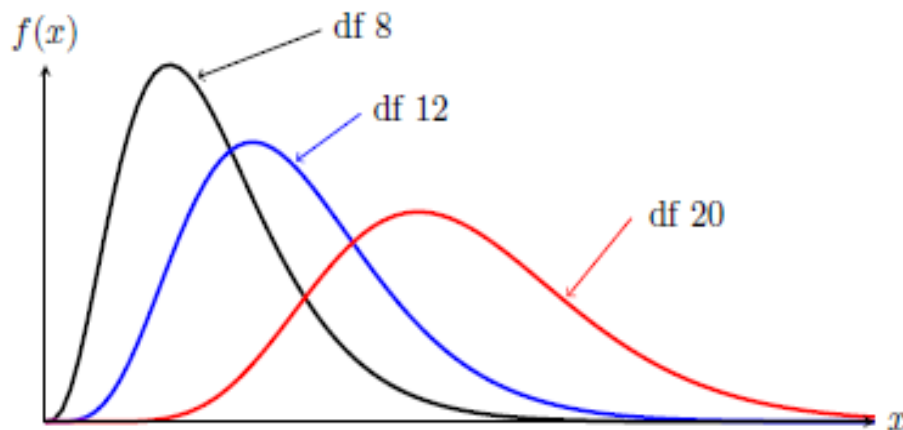
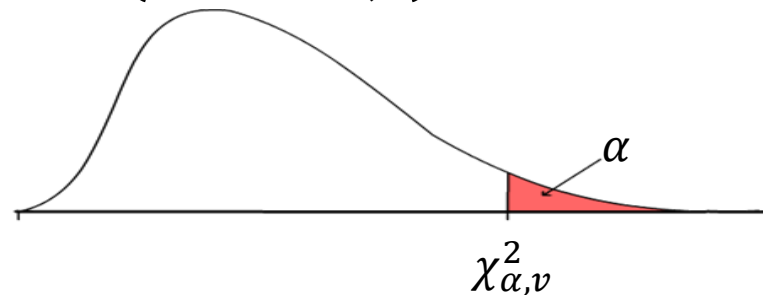


Table 5 Values of χ^2_{α}									
ν	$\alpha = 0.995$	$\alpha = 0.99$	$\alpha = 0.975$	$\alpha = 0.95$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$	$\alpha = 0.005$	ν
1	0.0000393	0.000157	0.000982	0.00993	3.841	5.024	6.635	7.879	1
2	0.0100	0.0201	0.0506	0.103	5.991	7.378	9.210	10.597	2
3	0.0717	0.115	0.216	0.352	7.815	9.348	11.345	12.838	3
4	0.207	0.297	0.484	0.711	9.488	11.143	13.277	14.860	4
5	0.412	0.554	0.831	1.145	11.070	12.833	15.086	16.750	5
6	0.676	0.872	1.237	1.635	12.592	14.449	16.812	18.548	6
7	0.989	1.239	1.690	2.167	14.067	16.013	18.475	20.278	7
8	1.344	1.646	2.180	2.733	15.507	17.535	20.090	21.955	8
9	1.735	2.088	2.700	3.325	16.919	19.023	21.666	23.589	9
10	2.156	2.558	3.247	3.940	18.307	20.483	23.209	25.188	10
11	2.603	3.053	3.816	4.575	19.675	21.920	24.725	26.757	11
12	3.074	3.571	4.404	5.226	21.026	23.337	26.217	28.300	12
13	3.565	4.107	5.009	5.892	22.362	24.736	27.688	29.819	13
14	4.075	4.660	5.629	6.571	23.685	26.119	29.141	31.319	14
15	4.601	5.229	6.262	7.261	24.996	27.488	30.578	32.801	15
16	5.142	5.812	6.908	7.962	26.296	28.845	32.000	34.267	16
17	5.697	6.408	7.564	8.672	27.587	30.191	33.409	35.718	17
18	6.265	7.015	8.231	9.390	28.869	31.526	34.805	37.156	18
19	6.844	7.633	8.907	10.117	30.144	32.852	36.191	38.582	19
20	7.434	8.260	9.591	10.851	31.410	34.170	37.566	39.997	20
21	8.034	8.897	10.283	11.591	32.671	35.479	38.932	41.401	21
22	8.643	9.542	10.982	12.338	33.924	36.781	40.289	42.796	22
23	9.260	10.196	11.689	13.091	35.172	38.076	41.638	44.181	23
24	9.886	10.856	12.401	13.848	36.415	39.364	42.980	45.559	24
25	10.520	11.524	13.120	14.611	37.652	40.646	44.314	46.928	25
26	11.160	12.198	13.844	15.379	38.885	41.923	45.642	48.290	26
27	11.808	12.879	14.573	16.151	40.113	43.195	46.963	49.645	27
28	12.461	13.563	15.308	16.928	41.337	44.461	48.278	50.993	28
29	13.121	14.256	16.047	17.708	42.557	45.722	49.588	52.336	29
30	13.787	14.953	16.791	18.493	43.773	46.979	50.892	53.672	30
40	20.707	22.164	24.433	26.509	55.758	59.342	63.691	66.766	40
50	27.991	29.707	32.357	34.764	67.505	71.420	76.154	79.490	50
60	35.534	37.485	40.482	43.188	79.082	83.298	88.379	91.952	60
70	43.275	45.442	48.758	51.739	90.531	95.023	100.425	104.215	70
80	51.172	53.540	57.153	60.391	101.879	106.629	112.329	116.321	80
90	59.196	61.754	65.647	69.126	113.145	118.136	124.116	128.299	90
100	67.328	70.065	74.222	77.929	124.342	129.561	135.807	140.169	100

Table A.5

$\chi^2_{\alpha,\nu}$: chi-squared critical value

$$p(\chi^2_{\nu} > \chi^2_{\alpha,\nu}) = \alpha$$



eg.

$$\chi^2_{0.025,16} = 28.845$$

Table A.5 Chi-square Distribution Table (1)

$\alpha = .10$	.05	.025	.01	.005	.001	df
2.706	3.841	5.024	6.635	7.879	10.83	1
4.605	5.991	7.378	9.210	10.60	13.82	2
6.251	7.815	9.348	11.34	12.84	16.27	3
7.779	9.488	11.14	13.28	14.86	18.47	4
9.236	11.07	12.83	15.09	16.75	20.52	5
10.64	12.59	14.45	16.81	18.55	22.46	6
12.02	14.07	16.01	18.48	20.28	24.32	7
13.36	15.51	17.53	20.09	21.95	26.12	8
14.68	16.92	19.02	21.67	23.59	27.88	9
15.99	18.31	20.48	23.21	25.19	29.59	10
17.28	19.68	21.92	24.72	26.76	31.27	11
18.55	21.03	23.34	26.22	28.30	32.91	12
19.81	22.36	24.74	27.69	29.82	34.53	13
21.06	23.68	26.12	29.14	31.32	36.12	14
22.31	25.00	27.49	30.58	32.80	37.70	15
23.54	26.30	28.85	32.00	34.27	39.25	16
24.77	27.59	30.19	33.41	35.72	40.79	17
25.99	28.87	31.53	34.81	37.16	42.31	18
27.20	30.14	32.85	36.19	38.58	43.82	19
28.41	31.41	34.17	37.57	40.00	45.31	20
29.62	32.67	35.48	38.93	41.40	46.80	21
30.81	33.92	36.78	40.29	42.80	48.27	22
32.01	35.17	38.08	41.64	44.18	49.73	23
33.20	36.42	39.36	42.98	45.56	51.18	24
34.38	37.65	40.65	44.31	46.93	52.62	25
35.56	38.89	41.92	45.64	48.29	54.05	26
36.74	40.11	43.19	46.96	49.65	55.48	27
37.92	41.34	44.46	48.28	50.99	56.89	28
39.09	42.56	45.72	49.59	52.34	58.30	29
40.26	43.77	46.98	50.89	53.67	59.70	30
51.81	55.76	59.34	63.69	66.77	73.40	40
63.17	67.50	71.42	76.15	79.49	86.66	50
74.40	79.08	83.30	88.38	91.95	99.61	60
85.53	90.53	95.02	100.43	104.21	112.32	70
96.58	101.88	106.63	112.33	116.32	124.84	80
107.57	113.15	118.14	124.12	128.30	137.21	90
118.50	124.34	129.56	135.81	140.17	149.45	100
140.23	146.57	152.21	158.95	163.65	173.62	120
268.47	277.14	284.80	293.89	300.18	313.44	240

Chi-square Distribution Table (2)

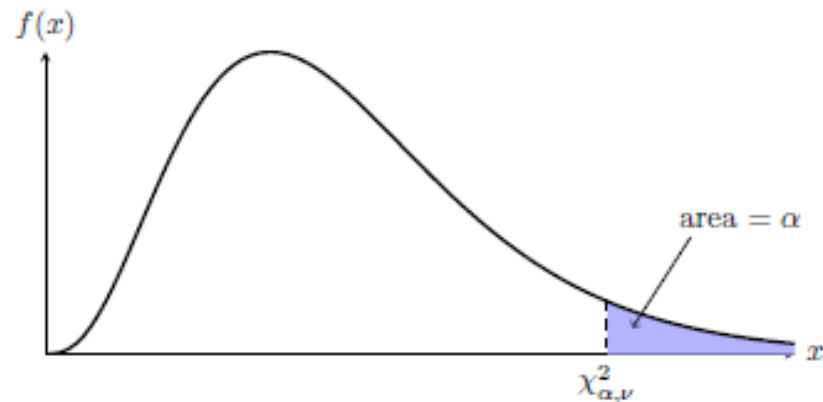
df	$\alpha =$	.999	.995	.99	.975	.95	.90
1		.000002	.000039	.000157	.000982	.003932	.01579
2		.002001	.01003	.02010	.05064	.1026	.2107
3		.02430	.07172	.1148	.2158	.3518	.5844
4		.09080	.2070	.2971	.4844	.7107	1.064
5		.2102	.4117	.5543	.8312	1.145	1.610
6		.3811	.6757	.8721	1.237	1.635	2.204
7		.5985	.9893	1.239	1.690	2.167	2.833
8		.8571	1.344	1.646	2.180	2.733	3.490
9		1.152	1.735	2.088	2.700	3.325	4.168
10		1.479	2.156	2.558	3.247	3.940	4.865
11		1.834	2.603	3.053	3.816	4.575	5.578
12		2.214	3.074	3.571	4.404	5.226	6.304
13		2.617	3.565	4.107	5.009	5.892	7.042
14		3.041	4.075	4.660	5.629	6.571	7.790
15		3.483	4.601	5.229	6.262	7.261	8.547
16		3.942	5.142	5.812	6.908	7.962	9.312
17		4.416	5.697	6.408	7.564	8.672	10.09
18		4.905	6.265	7.015	8.231	9.390	10.86
19		5.407	6.844	7.633	8.907	10.12	11.65
20		5.921	7.434	8.260	9.591	10.85	12.44
21		6.447	8.034	8.897	10.28	11.59	13.24
22		6.983	8.643	9.542	10.98	12.34	14.04
23		7.529	9.260	10.20	11.69	13.09	14.85
24		8.085	9.886	10.86	12.40	13.85	15.66
25		8.649	10.52	11.52	13.12	14.61	16.47
26		9.222	11.16	12.20	13.84	15.38	17.29
27		9.803	11.81	12.88	14.57	16.15	18.11
28		10.39	12.46	13.56	15.31	16.93	18.94
29		10.99	13.12	14.26	16.06	17.71	19.77
30		11.59	13.79	14.95	16.79	18.49	20.60
40		17.92	20.71	22.16	24.43	26.51	29.05
50		24.67	27.99	29.71	32.36	34.76	37.69
60		31.74	35.53	37.48	40.48	43.19	46.46
70		39.04	43.28	45.44	48.76	51.74	55.33
80		46.52	51.17	53.54	57.15	60.39	64.28
90		54.16	59.20	61.75	65.65	69.13	73.29
100		61.92	67.33	70.06	74.22	77.93	82.36
120		77.76	83.85	86.92	91.57	95.70	100.62
240		177.95	187.32	191.99	198.98	205.14	212.39

Using R,

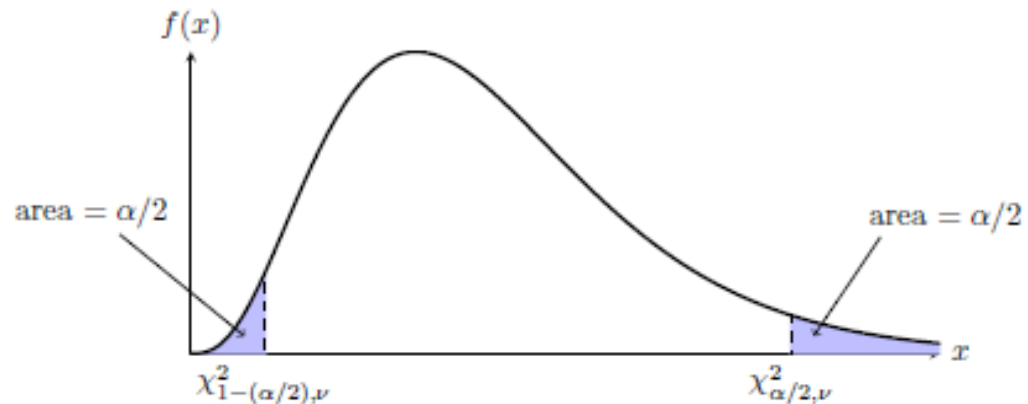
$P(X < c)$, where $X \sim \chi^2_v$:
>pchisq(c,v)

100p-th percentile:
>qchisq(p, v)

$\chi^2_{\alpha,v}$:
>qchisq(1- α , v)



$\chi^2_{1-\alpha,v}$:
>qchisq(α , v)



Example 6.10: The claim that $\sigma^2 = 21.3$ for a normal population is rejected if s^2 for a sample of $n = 15$ exceeds 39.74. What's the probability that the claim is rejected even though $\sigma^2 = 21.3$?

$$n = 15$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{14(39.74)}{21.3} = 26.12$$

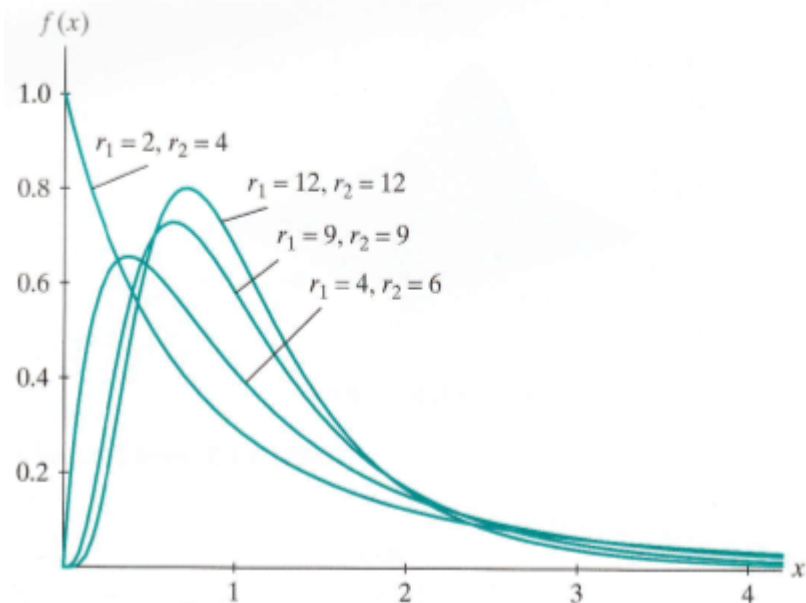
$$P(S^2 > 39.74) = P(\chi^2 > 26.12) = 0.025 \text{ because}$$
$$\chi_{0.025, n-1}^2 = \chi_{0.025, 14}^2 = 26.12.$$

Using R,
>1-pchisq(26.12, 14)

If S_1^2 & S_2^2 are the variances of independent random samples of size n_1 & n_2 , respectively, taken from two normal populations with the same variance, then

$$\frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$$

F_{r_1, r_2} density function



Note: If $F \sim F_{n_1-1, n_2-1}$, then $\frac{1}{F} \sim F_{n_2-1, n_1-1}$.

F_{α, v_1, v_2} : F critical value

$$P(F_{v_1, v_2} > F_{\alpha, v_1, v_2}) = \alpha$$

df ₂	α	df ₁									
		1	2	3	4	5	6	7	8	9	10
7	.25	1.57	1.70	1.72	1.72	1.71	1.71	1.70	1.70	1.69	1.69
	.10	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70
	.05	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
	.025	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76
	.01	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
	.005	16.24	12.40	10.88	10.05	9.52	9.16	8.89	8.68	8.51	8.38
	.001	29.25	21.69	18.77	17.20	16.21	15.52	15.02	14.63	14.33	14.08
8	.25	1.54	1.66	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63
	.10	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54
	.05	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
	.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
	.01	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
	.005	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21
	.001	25.41	18.49	15.83	14.39	13.48	12.86	12.40	12.05	11.77	11.54
9	.25	1.51	1.62	1.63	1.63	1.62	1.61	1.60	1.60	1.59	1.59
	.10	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42
	.05	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
	.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
	.01	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
	.005	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42
	.001	22.86	16.39	13.90	12.56	11.71	11.13	10.70	10.37	10.11	9.89
10	.25	1.49	1.60	1.60	1.59	1.59	1.58	1.57	1.56	1.56	1.55
	.10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32
	.05	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
	.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72
	.01	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85
	.005	12.83	9.43	8.08	7.34	6.87	6.54	6.30	6.12	5.97	5.85
	.001	21.04	14.91	12.55	11.28	10.48	9.93	9.52	9.20	8.96	8.75
11	.25	1.47	1.58	1.58	1.57	1.56	1.55	1.54	1.53	1.53	1.52
	.10	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25
	.05	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
	.025	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53
	.01	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54
	.005	12.23	8.91	7.60	6.88	6.42	6.10	5.86	5.68	5.54	5.42
	.001	19.69	13.81	11.56	10.35	9.58	9.05	8.66	8.35	8.12	7.92
12	.25	1.46	1.56	1.56	1.55	1.54	1.53	1.52	1.51	1.51	1.50
	.10	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19
	.05	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
	.025	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37
	.01	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30
	.005	11.75	8.51	7.23	6.52	6.07	5.76	5.52	5.35	5.20	5.09
	.001	18.64	12.97	10.80	9.63	8.89	8.38	8.00	7.71	7.48	7.29

df ₁											df ₂
12	15	20	24	30	40	60	120	240	inf.	α	
1.68	1.68	1.67	1.67	1.66	1.66	1.65	1.65	1.65	1.65	.25	7
2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.48	2.47	.10	
3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.25	3.23	.05	
4.67	4.57	4.47	4.41	4.36	4.31	4.25	4.20	4.17	4.14	.025	
6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.69	5.65	.01	
8.18	7.97	7.75	7.64	7.53	7.42	7.31	7.19	7.13	7.08	.005	
13.71	13.32	12.93	12.73	12.53	12.33	12.12	11.91	11.80	11.70	.001	
1.62	1.62	1.61	1.60	1.60	1.59	1.59	1.58	1.58	1.58	.25	8
2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.30	2.29	.10	
3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.95	2.93	.05	
4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.70	3.67	.025	
5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.90	4.86	.01	
7.01	6.81	6.61	6.50	6.40	6.29	6.18	6.06	6.01	5.95	.005	
11.19	10.84	10.48	10.30	10.11	9.92	9.73	9.53	9.43	9.33	.001	
1.58	1.57	1.56	1.56	1.55	1.54	1.64	1.53	1.53	1.53	.25	9
2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.17	2.16	.10	
3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.73	2.71	.05	
3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.36	3.33	.025	
5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.35	4.31	.01	
6.23	6.03	5.83	5.73	5.62	5.52	5.41	5.30	5.24	5.19	.005	
9.57	9.24	8.90	8.72	8.55	8.37	8.19	8.00	7.91	7.81	.001	
1.54	1.53	1.52	1.52	1.51	1.51	1.50	1.49	1.49	1.48	.25	10
2.28	2.24	2.20	2.18	2.16	2.13	2.11	2.08	2.07	2.06	.10	
2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.56	2.54	.05	
3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.11	3.08	.025	
4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.95	3.91	.01	
5.66	5.47	5.27	5.17	5.07	4.97	4.86	4.75	4.69	4.64	.005	
8.45	8.13	7.80	7.64	7.47	7.30	7.12	6.94	6.85	6.76	.001	
1.51	1.50	1.49	1.49	1.48	1.47	1.47	1.46	1.45	1.45	.25	11
2.21	2.17	2.12	2.10	2.08	2.05	2.03	2.00	1.99	1.97	.10	
2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.43	2.40	.05	
3.43	3.33	3.23	3.17	3.12	3.06	3.00	2.94	2.91	2.88	.025	
4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.65	3.60	.01	
5.24	5.05	4.86	4.76	4.65	4.55	4.45	4.34	4.28	4.23	.005	
7.63	7.32	7.01	6.85	6.68	6.52	6.35	6.18	6.09	6.00	.001	
1.49	1.48	1.47	1.46	1.45	1.45	1.44	1.43	1.43	1.42	.25	12
2.15	2.10	2.06	2.04	2.01	1.99	1.96	1.93	1.92	1.90	.10	
2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.32	2.30	.05	
3.28	3.18	3.07	3.02	2.96	2.91	2.85	2.79	2.76	2.72	.025	
4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.41	3.36	.01	
4.91	4.72	4.53	4.43	4.33	4.23	4.12	4.01	3.96	3.90	.005	
7.00	6.71	6.40	6.25	6.09	5.93	5.76	5.59	5.51	5.42	.001	

df ₁											df ₂
12	15	20	24	30	40	60	120	240	inf.	α	
1.40	1.38	1.37	1.36	1.35	1.34	1.33	1.32	1.31	1.30	.25	19
1.91	1.86	1.81	1.79	1.76	1.73	1.70	1.67	1.65	1.63	.10	
2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.90	1.88	.05	
2.72	2.62	2.51	2.45	2.39	2.33	2.27	2.20	2.17	2.13	.025	
3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.54	2.49	.01	
3.76	3.59	3.40	3.31	3.21	3.11	3.00	2.89	2.83	2.78	.005	
4.97	4.70	4.43	4.29	4.14	3.99	3.84	3.68	3.60	3.51	.001	
1.39	1.37	1.36	1.35	1.34	1.33	1.32	1.31	1.30	1.29	.25	20
1.89	1.84	1.79	1.77	1.74	1.71	1.68	1.64	1.63	1.61	.10	
2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.87	1.84	.05	
2.68	2.57	2.46	2.41	2.35	2.29	2.22	2.16	2.12	2.09	.025	
3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.47	2.42	.01	
3.68	3.50	3.32	3.22	3.12	3.02	2.92	2.81	2.75	2.69	.005	
4.82	4.56	4.29	4.15	4.00	3.86	3.70	3.54	3.46	3.38	.001	
1.38	1.37	1.35	1.34	1.33	1.32	1.31	1.30	1.29	1.28	.25	21
1.87	1.83	1.78	1.75	1.72	1.69	1.66	1.62	1.60	1.59	.10	
2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.84	1.81	.05	
2.64	2.53	2.42	2.37	2.31	2.25	2.18	2.11	2.08	2.04	.025	
3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.41	2.36	.01	
3.60	3.43	3.24	3.15	3.05	2.95	2.84	2.73	2.67	2.61	.005	
4.70	4.44	4.17	4.03	3.88	3.74	3.58	3.42	3.34	3.26	.001	
1.37	1.36	1.34	1.33	1.32	1.31	1.30	1.29	1.28	1.28	.25	22
1.86	1.81	1.76	1.73	1.70	1.67	1.64	1.60	1.59	1.57	.10	
2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.81	1.78	.05	
2.60	2.50	2.39	2.33	2.27	2.21	2.14	2.08	2.04	2.00	.025	
3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.35	2.31	.01	
3.54	3.36	3.18	3.08	2.98	2.88	2.77	2.66	2.60	2.55	.005	
4.58	4.33	4.06	3.92	3.78	3.63	3.48	3.32	3.23	3.15	.001	
1.37	1.35	1.34	1.33	1.32	1.31	1.30	1.28	1.28	1.27	.25	23
1.84	1.80	1.74	1.72	1.69	1.66	1.62	1.59	1.57	1.55	.10	
2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.79	1.76	.05	
2.57	2.47	2.36	2.30	2.24	2.18	2.11	2.04	2.01	1.97	.025	
3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.31	2.26	.01	
3.47	3.30	3.12	3.02	2.92	2.82	2.71	2.60	2.54	2.48	.005	
4.48	4.23	3.96	3.82	3.68	3.53	3.38	3.22	3.14	3.05	.001	
1.36	1.35	1.33	1.32	1.31	1.30	1.29	1.28	1.27	1.26	.25	24
1.83	1.78	1.73	1.70	1.67	1.64	1.61	1.57	1.55	1.53	.10	
2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.76	1.73	.05	
2.54	2.44	2.33	2.27	2.21	2.15	2.08	2.01	1.97	1.94	.025	
3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.26	2.21	.01	
3.42	3.25	3.06	2.97	2.87	2.77	2.66	2.55	2.49	2.43	.005	
4.39	4.14	3.87	3.74	3.59	3.45	3.29	3.14	3.05	2.97	.001	

Using R,

$P(X < c)$, where $X \sim F_{\nu_1, \nu_2}$:
>pf(c, ν_1 , ν_2)

100 p -th percentile:
>qf(p, ν_1 , ν_2)

F_{α, ν_1, ν_2} :
>qf(1- α , ν_1 , ν_2)

$F_{1-\alpha, \nu_1, \nu_2}$:
>qf(α , ν_1 , ν_2)

Example 6.11: Let $X_1, X_2, \dots, X_{10}$: random sample from $N(\mu_1, \sigma^2)$ independent with $Y_1, Y_2, \dots, Y_{13}$: random sample from $N(\mu_2, \sigma^2)$.

S_X^2 : sample variance of $X_1, X_2, \dots, X_{10}$

S_Y^2 : sample variance of $Y_1, Y_2, \dots, Y_{13}$

Find $P(S_X^2 > 2.8S_Y^2)$.

$$\frac{S_X^2}{S_Y^2} \sim F_{v_1, v_2}, \quad v_1 = n_1 - 1 = 9,$$

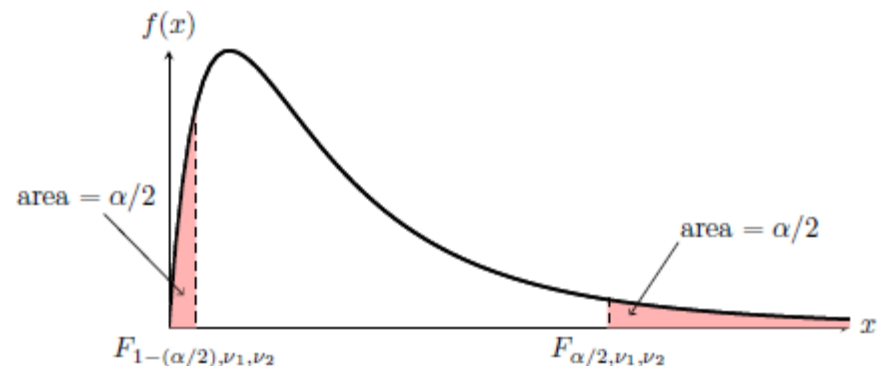
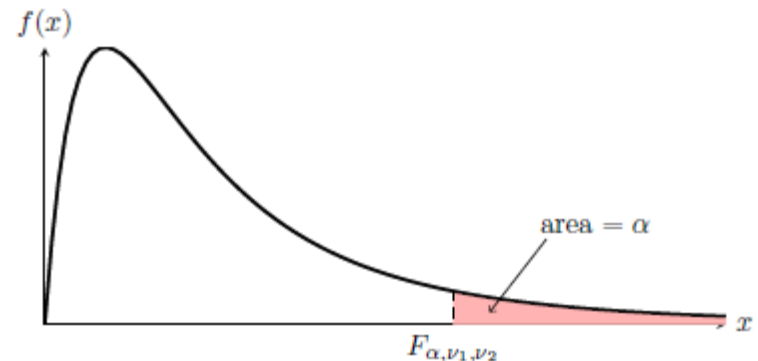
$$v_2 = n_2 - 1 = 12$$

$$F_{0.05, 9, 12} = 2.80 \text{ from Table A.6}$$

$$\begin{aligned} P(S_X^2 > 2.8S_Y^2) &= P\left(\frac{S_X^2}{S_Y^2} > 2.8\right) \\ &= P(F_{9, 12} > 2.8) = 0.05 \end{aligned}$$

Using R,

$$>1-\text{pf}(2.8, 9, 12)$$



Reciprocal Property of the F Distribution

$$\frac{S_X^2}{S_Y^2} \sim F_{\nu_1, \nu_2} \Rightarrow \frac{S_Y^2}{S_X^2} \sim F_{\nu_2, \nu_1}$$

$$\alpha = P\left(\frac{S_X^2}{S_Y^2} > F_{\alpha, \nu_1, \nu_2}\right) = P\left(\frac{S_X^2}{S_Y^2} > \frac{1}{F_{1-\alpha, \nu_2, \nu_1}}\right) = P\left(\frac{S_Y^2}{S_X^2} < F_{1-\alpha, \nu_2, \nu_1}\right)$$

Example 6.12:

$$F_{.05, 10, 7} = 3.64$$

$$F_{.95, 7, 10} = \frac{1}{F_{.05, 10, 7}} = \frac{1}{3.64} = 0.27$$